

# DesignDiffusion: High-Quality Text-to-Design Image Generation with Diffusion Models

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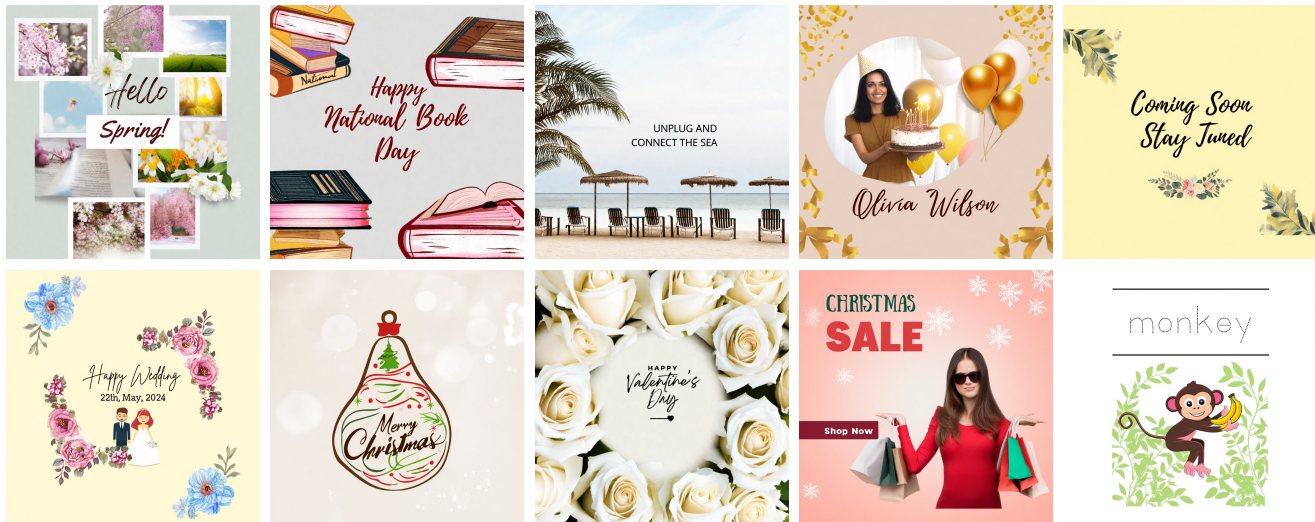


Figure 1. Images generated by our DesignDiffusion model, which only requires a simple text prompt as input and can generate diverse, high-quality design images with accurate textual and vivid visual content.

## Abstract

In this paper, we present DesignDiffusion, a simple yet effective framework for the novel task of synthesizing design images from textual descriptions. A primary challenge lies in generating accurate and style-consistent textual and visual content. Existing works in a related task of visual text generation often focus on generating text within given specific regions, which limits the creativity of generation models, resulting in style or color inconsistencies between textual and visual elements if applied to design image generation. To address this issue, we propose an end-to-end, one-stage diffusion-based framework that avoids intricate components like position and layout modeling. Specifically, the proposed framework directly synthesizes textual and visual design elements from user prompts. It utilizes a distinctive character embedding derived from the visual text to enhance the input prompt, along with a character localization loss for enhanced supervision during text generation. Furthermore, we employ a self-play Direct Preference Optimization fine-

tuning strategy to improve the quality and accuracy of the synthesized visual text. Extensive experiments demonstrate that DesignDiffusion achieves state-of-the-art performance in design image generation.

## 1. Introduction

Design images play a pivotal role in various applications such as graphic design, advertising, and scientific visualization, where the integration of visual and textual elements must be flawless.

Recently many fascinating and wonderful applications, such as art image creation or film production, have shown remarkable progress due to the rapid advance of text-to-image generation techniques [37–40, 47]. Despite these advancements, the precise generation of design images poses a distinct challenge, largely due to the limited capabilities of existing image generation models in complex layout planning and visual text generation.

To address the critical issue of visual text generation, re-

cent studies [27, 41] have made significant contributions. A common approach involves a two-step process: first generating an image from textual input and then identifying a suitable location within the image to insert text. However, this method presents several challenges. Firstly, a generated image may not have a suitable region for adding text, resulting in an overlay of text on visual elements. Secondly, even with a plausible location, the rendered text may lack consistency with the visual elements presented in the image, compromising overall coherence and professionalism.

An alternative research direction in visual text rendering, as suggested by [5, 6], focuses on learning the layout of text regions from textual prompts in advance. Then, this learned layout serves as a condition to guide text-to-image models in generating text within designated regions of the image. However, the dependence on predefined text regions might constrain the creative breadth of the generated images. Moreover, these predetermined text regions may not always adhere to natural design principles, potentially leading to visually unattractive or conceptually incoherent outputs.

To address the aforementioned issues, we propose DesignDiffusion, an end-to-end diffusion-based framework specifically developed for the novel task of generating design images from text. The holistic learning approach presents multiple benefits: (1) it eliminates the need to predetermine text regions in the generative images, thereby preserving the creative freedom of the generative models; (2) it enables the smooth incorporation of generated text into images, yielding cohesive designs that accurately reflect input prompts.

In our framework, we employ three essential techniques when fine-tuning text-to-image diffusion models: a) Recognizing that a conventional CLIP text encoder might overlook the specific textual content designated for rendering in the generated image, we propose a character-level decomposition of words. This approach provides the CLIP text encoder hints to render each character in the image precisely. b) To encourage the model to pay more attention to newly introduced character embeddings, we propose a character localization loss. This loss function compels each character embedding to concentrate on its respective area within the image, ensuring an accurate visual representation of each character. c) We introduce a new training strategy called Self-Play Direct Preference Optimization (SP-DPO) for fine-tuning the model to enhance its capability of generating accurate and high-fidelity text. SP-DPO is based on the hypothesis that ground truth data are typically preferred by humans over model-generated samples. Based on this, we fine-tune the model to align with these human preferences.

Extensive experiments demonstrate the superior capabilities of DesignDiffusion for design image generation. It generates creative design images with prompts, as shown in Fig. 1. Quantitative and qualitative analysis shows that our approach significantly surpasses current state-of-the-art

text-to-image and visual text rendering models in terms of both image quality and text accuracy.

In summary, our contributions are presented as follows:

- We propose DesignDiffusion, an end-to-end, diffusion-based framework for text-to-design image generation, which enables the simultaneous generation of image and visual text elements, eliminating the necessity for predefined text regions or traditional two-stage separated text and image creation process.
- We explore prompt enhancement for CLIP text encoding along with a character localization loss, to improve the accuracy of text placement and the fidelity of text rendering.
- By adopting an SP-DPO mechanism, the generative model is further fine-tuned, achieving remarkable improvements in visual text accuracy and overall image quality.

## 2. Related Work

**Visual Text Rendering.** Despite the remarkable success of existing diffusion models [37, 38, 40] in text-to-image generation, visual text rendering remains a challenging task. Recently, researchers have recognized this issue and proposed several approaches to address it, such as GlyphDraw [29], DiffUTE [4], GlyphControl [45], AnyText [41], TextDiffuser series [5, 6], and [26]. While these efforts have improved text rendering accuracy, it is worth noting that these models obtain limited image quality due to the need of pre-giving text rendering areas during the generation process or inpainting text into a well-generated image. Separating text and visual content generation away seriously limits the overall quality of generated images with visual text rendered, which is incompatible with the goal of design image generation.

**Compositional Image Generation.** To improve the ability of text-to-image models to create complex compositions, researchers have integrated new components into diffusion models, enabling the generation of images with multiple features. Examples include T2I-Adapter [30] and ControlNet [47], which incorporate semantic details like layout and pose to influence the structure of images [48]. In a similar vein, GLIGEN [21] utilizes adapters with diffusion models to achieve image generation that is conditioned on specific spatial arrangements. However, these methods cannot generate images directly from prompts and require substantial human effort to define the image layout. On the other hand, some studies have investigated methods that do not require additional training to prompt diffusion models to create images based on spatial conditions during the inference phase. Recent advancements [22, 32, 44] have seen large language models aiding diffusion models in generating images with intricate or predefined layouts. Yet, these approaches often fall short in generating design-centric images that seamlessly integrate both textual and visual components in a well-organized manner.

**Aligning Diffusion Models.** Replicating the success of re-

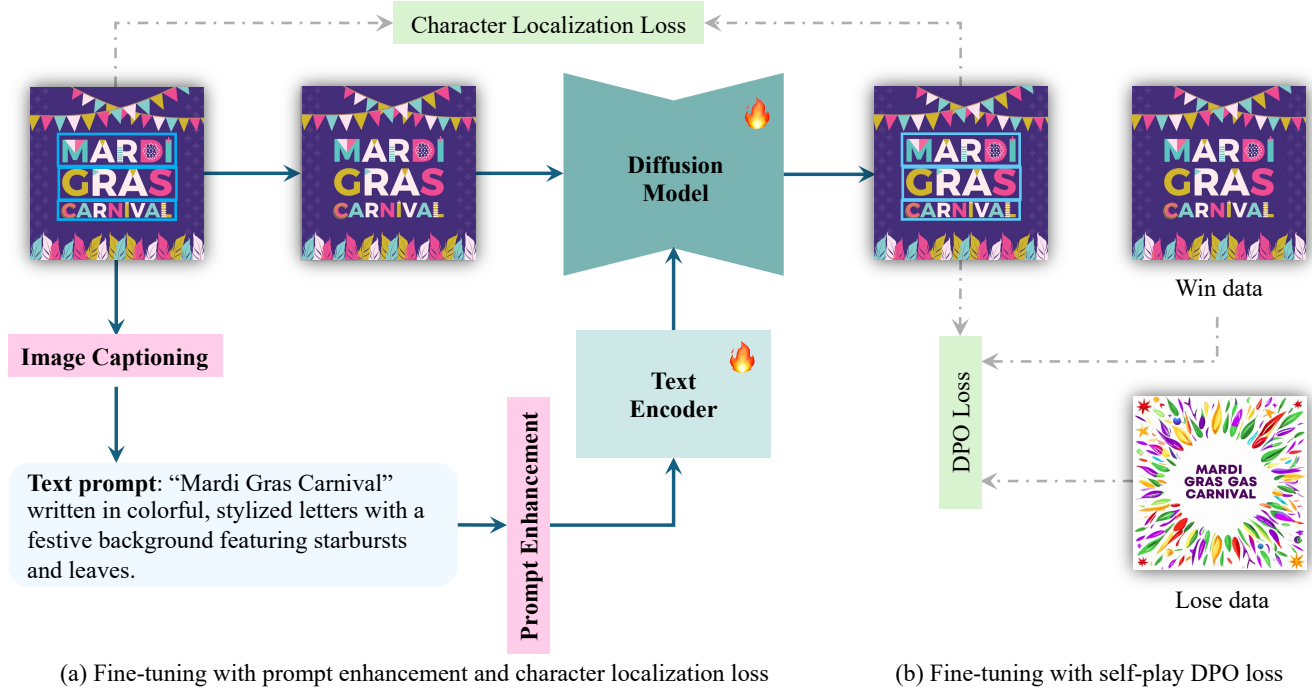


Figure 2. Overview of the DesignDiffusion framework. DesignDiffusion is based on enhanced text prompts, with trainable CLIP text encoder and UNet, and does not require additional complex conditions (glyphs, positions). Character localization loss is added as extra supervision at cross-attention maps to force the UNet to attend more to the visual character regions. To further improve the quality of visual text generation, we incorporate a self-play DPO strategy into the fine-tuning process. Diffusion denoise loss is omitted here.

inforcement learning in large language models [31, 35] to text-to-image diffusion models remains a significant challenge. Recent works have proposed various methods to address this problem. For example, EMU [7] identifies that a few high-quality samples are crucial when fine-tuning the model to generate visually appealing images. DPOK [12] and DDPO [3] use RL-based approaches to maximize rewards on a limited vocabulary set, performing well when the number of train/test prompts is small. Another work DiffusionDPO [42] applies DPO based on a human-annotated preference dataset. Our DesignDiffusion differs by applying a self-play DPO strategy, which does not require human annotation and lacks of exploration in design image generation.

### 3. DesignDiffusion

The generative community has witnessed the strong generative capability of diffusion models [9, 16, 34, 38, 48]. However, generating design images remains a great challenge, particularly in the context of exquisite layout and text rendering. Previous approaches [5, 6] in a related task (visual text rendering) attempt to inpaint text after generating an image, or to plan text regions using large language models such as Vicuna [49] and GPT [1] before generating an image. We argue that they do not consider the integration of text and image content. Note that, the text and image contexts in design images are often intertwined and inseparable, such

that they are incompatible with design image generation.

In contrast, we introduce a straightforward, one-stage approach for text-to-design image generation, relying solely on the user prompt. It is based on an aesthetic model, *i.e.*, Stable Diffusion XL (SDXL) model [34], incorporating three core components that enable the successful generation of text-rich design images. An overview of the DesignDiffusion framework is provided in Fig. 2. The DesignDiffusion training involves a two-stage fine-tuning strategy. In the first stage, we introduce prompt enhancement and character localization loss into the framework to generate high-quality images with accurate text. In the second stage, we propose SP-DPO to further improve the text rendering accuracy. These components are detailed in the following subsections.

#### 3.1. Prompt Enhancement for CLIP Text Encoding

Currently, available text-to-image diffusion models struggle to generate text-rich design images based solely on an input text prompt. This challenge arises from *two issues*: 1) Visually rendered text must be letter-by-letter correct, however, during tokenization, most words are tokenized as a single token, ignoring each letter within the word thus complicating the learning of individual letters; 2) Visual text in images is often diverse in style, size, and layout, increasing the complexity of visual text modeling.

To address the first issue, we introduce a novel approach

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**Algorithm 1** Self-play DPO fine-tuning, given a base diffusion model  $\epsilon_{\text{ref}}$ , a trainable copy  $\epsilon_{\theta}$ , a prompt-image paired design-image dataset  $\mathbf{D} = \{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^N$ , in which  $\mathbf{x}_i$  and  $\mathbf{y}_i$  represent image and prompt, respectively.

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for all  $i$  in  $\{1, \dots, N\}$  do
  Generate  $K$  candidate samples  $\{g_{ik}\}_{k=1}^K$  using  $\epsilon_{\text{ref}}$  with input  $\mathbf{y}_i$ 
  Losing data:  $\mathbf{x}_i^l = g_{ik}$ , in which  $g_{ik}$  gets the worst text rendering accuracy
  Winning data:  $\mathbf{x}_i^w = \mathbf{x}_i$ 
  Update  $\epsilon_{\theta}$  by:  $\min_{\epsilon_{\theta}} \sum_{i=0}^n (\log \sigma(-\beta T \omega(\lambda_t)) (\|\epsilon_i^w - \epsilon_{\theta}(\mathbf{x}_{i,t}^w, \mathbf{y}_i, t_i)\|_2^2 - \|\epsilon_i^w - \epsilon_{\text{ref}}(\mathbf{x}_{i,t}^w, \mathbf{y}_i, t_i)\|_2^2 - (\|\epsilon_i^l - \epsilon_{\theta}(\mathbf{x}_{i,t}^l, \mathbf{y}_i, t_i)\|_2^2 - \|\epsilon_i^l - \epsilon_{\text{ref}}(\mathbf{x}_{i,t}^l, \mathbf{y}_i, t_i)\|_2^2)))$ 
end for
return  $\epsilon_{\theta}$ 

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by embedding each letter from prompts directly into CLIP text encoders within latent diffusion models [34, 38]. Specifically, the input prompt of the CLIP text encoders is enhanced by appending an extra text description to the original prompt. The template for the text description is “Rendered characters: <startofchar> ... <endofchar>”. For example, as illustrated in Fig. 2, if the desired rendered text is “Mardi Gras Carnival”, the appended text description would be “Rendered characters:<startofchar> <M|> <la|> <lr|> <ld|> <li|><endofchar><startofchar> <I|G|> <lr|> <la|> <ls|> <endofchar> <startofchar> <I|C|> <la|> <lr|> <ln|> <li|> <lv|> <la|> <ll|><endofchar>”. A character surrounded by ‘|’ denotes a newly introduced trainable token.

We introduce a total of 97 new tokens, encompassing 26 uppercase letters, 26 lowercase letters, 10 numbers, 32 punctuation marks, a space, a start flag, and an end flag. The enhanced prompt enables the CLIP text encoder to be trained specifically to learn hints for rendering text. By augmenting the prompt with detailed rendered text information, the UNet learns to generate images that include the specified visual text content seamlessly integrated into the image, without relying on predefined text glyphs or positions. This approach addresses the challenge of integrating text and image content in a tightly coupled manner, distinguishing it from previous approaches that treated text and image generation as separate stages. This integrated technique ensures that design images with visual text are generated in a more natural and contextually appropriate way.

### 3.2. Localizing Cross-Attention Maps with Character Segmentation Masks

To address the second issue, *i.e.*, complexity (*e.g.*, style, size, color, and orientation) of text in images, we explore applying a text localization loss during training. This approach helps the model capture text effectively. Previous works [13, 43] have shown that cross-attention maps in the UNet component of latent diffusion models [34, 38] control which pixels attend to which tokens of the prompt text during diffusion generation. In our framework, the newly introduced trainable character embeddings in CLIP text encoders are not

involved in the original UNet pre-training. Consequently, the cross-attention maps between rendered character tokens and latent pixel features may be unreasonable. To encourage UNet to attend to the newly introduced character embeddings, we propose a character localization loss to associate each character token with its corresponding visual character region. Let  $\mathbf{A}^{(h \times w \times c)}$  denote the cross-attention maps, where  $h$  and  $w$  are the spatial dimensions of the latent in Stable Diffusion, respectively.  $\mathbf{A}^i = \mathbf{A}[:, :, i]$  represents the relationship between the image features and the  $i$ -th prompt token.  $\mathbf{M}^i$  is the region mask of the  $i$ -th prompt token, with 1 denoting the existence of elements. Encouraging each character token to attend to the corresponding character region, the character localization loss is formulated as follows:

$$\mathcal{L}_{\text{char}} = -\frac{1}{c} \sum_{i=0}^c (\mathbf{A}^i * \mathbf{M}^i - \mathbf{A}^i * (1 - \mathbf{M}^i)). \quad (1)$$

This character localization loss is used as a regularization term that forces the generation of visual characters to attend to the corresponding character embedding. We use the cross-attention maps in the five middle stages of UNet to calculate  $\mathcal{L}_{\text{char}}$  in our experiments.

### 3.3. Self-Play DPO Fine-Tuning for More Accurate Text Rendering

Recent diffusion DPO [36] is an alternative to RLHF to steer the model to the preferred distribution (winning data) and push it away from the unpreferred distribution (losing data). To further enhance the model’s ability to handle design images, we apply self-play DPO fine-tuning with our training GT images as winning data and synthetic data as losing data. Specifically, we generate losing data from the preliminary fine-tuned model in the first stage. To avoid high-quality generated images being mistakenly regarded as losing samples, we generate  $K$  samples for each prompt. Subsequently, we use a large multi-modality model [23, 24] to extract the visual text in images. The images with the poorest rendering quality are retained as the final losing data. The DPO reward objective is defined as follows:

$$L_{\text{DPO}}(\theta) = -\mathbb{E}_{c, \mathbf{x}_0^w, \mathbf{x}_0^l} [\log \sigma(\beta \log \frac{p_\theta(\mathbf{x}_0^w|c)}{p_{\text{ref}}(\mathbf{x}_0^w|c)} - \beta \log \frac{p_\theta(\mathbf{x}_0^l|c)}{p_{\text{ref}}(\mathbf{x}_0^l|c)})], \quad (2)$$

where  $c$  represents the condition,  $\mathbf{x}_0^w$  and  $\mathbf{x}_0^l$  denote the winning and losing data, respectively.  $\beta$  is a hyperparameter.  $p_{\text{ref}}$  is a reference distribution, and  $p_\theta$  is the optimized distribution with network parameters  $\theta$ .

Applying this loss to diffusion,  $p_\theta(\mathbf{x}_0|c)$  is not tractable, as it needs to marginalize out all possible diffusion paths  $(\mathbf{x}_1, \dots, \mathbf{x}_T)$  which leads to  $\mathbf{x}_0$ . So we utilize the evidence lower bound (ELBO) to overcome this challenge. Specifically, we need to redefine the reward function on the whole chain  $(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_T)$ . The Eq. 2 becomes to Eq. 3:

$$L_{\text{Diffusion-DPO}}(\theta) = -E_{x_0^w, x_0^l} \log \sigma(\beta E_{x_{0:T}^w, x_{0:T}^l} [\log \frac{p_\theta(x_{0:T}^w)}{p_{\text{ref}}(x_{0:T}^w)} - \log \frac{p_\theta(x_{0:T}^l)}{p_{\text{ref}}(x_{0:T}^l)}]), \quad (3)$$

$c$  is omitted for simplification. Since  $x_{1:T} \sim p_\theta(x_{1:T}|\mathbf{x}_0)$ ,  $T$  is usually large, and  $p_\theta(x_{1:T}|\mathbf{x}_0) = p_T \prod_{t=1}^T p_\theta(x_{t-1}|x_t)$ . Considering Jensen's inequality and convexity, we can further get the upper bound of Eq. 3:

$$L_{\text{Diffusion-DPO}}(\theta) < -E_{x_0^w, x_0^l, x_{t-1}^w \sim p_\theta(x_{t-1}^w|\mathbf{x}_0^w), x_{t-1}^l \sim p_\theta(x_{t-1}^l|\mathbf{x}_0^l)} \log \sigma(\beta T \log \frac{p_\theta(x_{t-1}^w|x_t^w)}{p_{\text{ref}}(x_{t-1}^w|x_t^w)} - \beta T \log \frac{p_\theta(x_{t-1}^l|x_t^l)}{p_{\text{ref}}(x_{t-1}^l|x_t^l)}). \quad (4)$$

Approximate the reverse process  $p_\theta(x_{1:T}|\mathbf{x}_0)$  with the forward process  $q(x_{1:T}|\mathbf{x}_0)$ , Eq. 4 becomes:

$$\begin{aligned} L(\theta) = & -E_{x_0^w, x_0^l, x_t^w \sim q(x_t^w|\mathbf{x}_0^w), x_t^l \sim q(x_t^l|\mathbf{x}_0^l)} \log \sigma(\beta T ( \\ & D_{\text{KL}}(q(x_{t-1}^w|x_{0,t}^w) || p_\theta(x_{t-1}^w|x_t^w)) \\ & - D_{\text{KL}}(q(x_{t-1}^w|x_{0,t}^w) || p_{\text{ref}}(x_{t-1}^w|x_t^w)) \\ & - D_{\text{KL}}(q(x_{t-1}^l|x_{0,t}^l) || p_\theta(x_{t-1}^l|x_t^l)) \\ & + D_{\text{KL}}(q(x_{t-1}^l|x_{0,t}^l) || p_{\text{ref}}(x_{t-1}^l|x_t^l)))). \end{aligned} \quad (5)$$

Given the diffusion reverse process, the final objective is:

$$\begin{aligned} L(\theta) = & -E_{x_0^w, x_0^l, x_t^w \sim q(x_t^w|\mathbf{x}_0^w), x_t^l \sim q(x_t^l|\mathbf{x}_0^l)} \\ & \log \sigma(-\beta T \omega(\lambda_t) (\|\epsilon^w - \epsilon_\theta(x_t^w, t)\|_2^2 \\ & - \|\epsilon^w - \epsilon_{\text{ref}}(x_t^w, t)\|_2^2 - \|\epsilon^l - \epsilon_\theta(x_t^l, t)\|_2^2 \\ & + \|\epsilon^l - \epsilon_{\text{ref}}(x_t^l, t)\|_2^2)), \end{aligned} \quad (6)$$

in which  $\omega(\lambda_t)$  is a timestep-wise weighting function, which is constant in common practice. Finally, the overall self-play DPO fine-tuning algorithm is described in Algorithm 1.

## 4. Experiments

### 4.1. Settings

**Dataset.** Due to the lack of high-quality design image dataset, we collect a dataset of 1 million high-quality design

images, including logos, posters, flyers, covers, etc. The collected dataset contains region masks that denote image layer information and text annotations including text content and font size in images. For image captioning, we use a large vision-language model LLaVA 1.6-34B [23–25] with improved reasoning, OCR, and world knowledge. We apply a pre-trained character segmentation model [5] to obtain character segmentation images. Considering text annotation and connected graph analysis, final character-wise region masks are obtained. In summary, our training dataset contains 1 million high-quality design images with caption, text, and character region annotations. Our testing dataset comprises 5k samples, kept unseen during training. For more details about the dataset, please refer to Appendix.

**Training details.** Our DesignDiffusion is trained using AdamW [28] with a learning rate of 1e-5. Xformers [20] and DeepSpeed [33] are applied during training to save memory further. Our model is trained on 4×NVIDIA A100 GPUs with a batch size of 256 in all our experiments. The coefficient  $\lambda$  for  $\mathcal{L}_{\text{char}}$  is 0.05 by default. After that, for DPO fine-tuning, we use 1.5k win-lose pairs as training data.  $\beta$  of DPO and  $T$  in Algorithm 1 are set as 5 and 1,000, respectively. The learning rate is 1e-8 and the number of candidate samples  $K = 10$ . Classifier-free guidance [15] is applied with a probability of 10% during training. When inference, the classifier-free guidance scale is set to 7.5 by default.

**Evaluation metrics.** Text-to-design image generation evaluation focuses on both image quality and text quality evaluation. (1) We apply **FID** [14] to evaluate the generated image distribution and the real design image distribution. (2) For evaluating the text quality in generated design images, we report **OCR** precision, recall, F-Score, and accuracy. Please refer to Appendix for a more detailed analysis of OCR tools. (3) Besides the evaluation from available tools or metrics, **human evaluation** is used in our experiments to evaluate the generated image and text quality comprehensively.

### 4.2. Comparison with State-of-the-Art Methods

**Quantitative Results.** We report the comparisons of the performance of design image generation in Tab. 1. We compare our method with SDXL [34], SD3 [11], FLUX.1 [19], DeepFloyd-IF [8], GlyphControl [46], Textdiffuser [5], Textdiffuser-2 [6], and AnyText [41]. The former four methods are open-source state-of-the-art text-to-image generation models. The latter three are state-of-the-art visual rendering or inpainting methods. Our DesignDiffusion outperforms other methods by a large margin with an FID score of 19.87, demonstrating the high quality of our generated design images. From the aspect of visual text quality, we report the OCR text precision, recall, F-score, and accuracy systematically and comprehensively. The comparison tells us that our DesignDiffusion gets a state-of-the-art visual text generation performance. The performance of our method in both

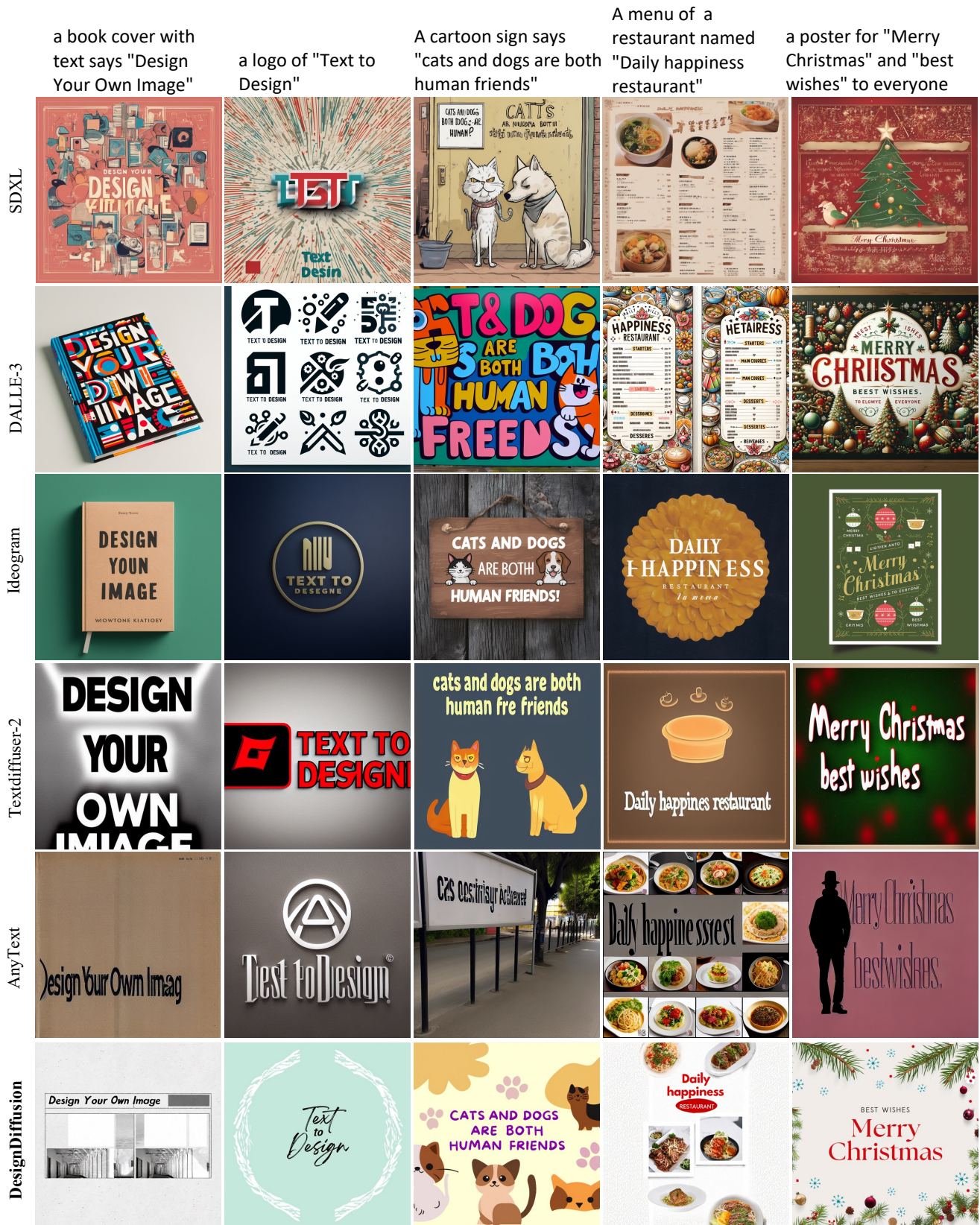


Figure 3. Qualitative comparisons with previous state-of-the-art text-to-image generation and visual text rendering methods reveal that our DesignDiffusion produces more elegant and harmonious integrated visual and textual design images.

Table 1. Quantitative comparisons with state-of-the-art text-to-image generation and visual text generation methods demonstrate that DesignDiffusion achieves superior performance in terms of FID, text precision, recall, F-score, and accuracy.

| Method                 | FID↓         | Text Precision↑ | Text Recall↑ | Text F-score↑ | Text Accuracy↑ |
|------------------------|--------------|-----------------|--------------|---------------|----------------|
| SDXL [34]              | 45.10        | 0.517           | 0.461        | 0.488         | 0.241          |
| SD3 [11]               | 45.90        | 0.754           | 0.675        | 0.712         | 0.415          |
| FLUX [19]              | 35.28        | 0.764           | 0.698        | 0.729         | 0.442          |
| DeepFloyd-IF [8]       | 52.05        | 0.607           | 0.534        | 0.568         | 0.290          |
| GlyphControl [46]      | 60.07        | 0.462           | 0.352        | 0.400         | 0.218          |
| Textdiffuser [5]       | 55.83        | 0.806           | 0.706        | 0.752         | 0.478          |
| Textdiffuser-2 [6]     | 63.92        | 0.866           | 0.792        | 0.827         | 0.583          |
| AnyText [41]           | 64.69        | 0.440           | 0.378        | 0.407         | 0.230          |
| <b>DesignDiffusion</b> | <b>19.87</b> | <b>0.888</b>    | <b>0.837</b> | <b>0.862</b>  | <b>0.631</b>   |

Table 2. Results of a user study comparing our method with previous approaches based on four aspects of text and image quality.

| Method             | Image Aesthetic↑ | Text Quality ↑ | Layout Aesthetics↑ | Text-Image Matching↑ |
|--------------------|------------------|----------------|--------------------|----------------------|
| SDXL [34]          | <b>0.43</b>      | 0.11           | 0.18               | 0.28                 |
| Textdiffuser-2 [6] | 0.08             | 0.34           | 0.22               | 0.14                 |
| AnyText [41]       | 0.09             | 0.03           | 0.07               | 0.06                 |
| DesignDiffusion    | 0.40             | <b>0.51</b>    | <b>0.54</b>        | <b>0.52</b>          |

the image and text metrics indicates that DesignDiffusion establishes a new baseline in design image generation.

**Qualitative Results.** We compare our method with state-of-the-art text-to-image models, *i.e.*, SDXL [34], DALLÉ-3 [2], Ideogram [17], and visual text rendering models, *i.e.*, Textdiffuser-2 [6], AnyText [41], as shown in Fig. 3. We observe that although state-of-the-art text-to-image models (SDXL, DALLÉ-3, Ideogram) can generate visually compelling images, they still struggle to generate accurate visual text. As for visual text rendering methods, we figure out that they generate visual text while brushing out the image content or reducing the aesthetics of the image seriously. It may be because that Textdiffuser-2 [6], AnyText [41] both rely on the text layout planning or strong text position fitting, leading to overfitting on visual text and overlooking other visual contents. Over-strong limitation of visual text area or other information limits the creativity of generating visual content. Throughout the comparisons, design images generated by our DesignDiffusion are more accurate than those from state-of-the-art text-to-image models and more elegant than those generated from visual text rendering models.

**User Studies.** We randomly generate 20 samples without any cherry pick compared with three methods, *i.e.*, SDXL [34], Textdiffuser-2 [6], and AnyText [41]. The participants were asked four questions about various aspects of image and text evaluation, in terms of image aesthetic, visual text quality, layout aesthetics, and text-image matching. We collect about 20 results from participants and report the preference ratio of the four metrics in Tab. 2. From the aspects of text quality, text layout aesthetic, and text-image matching, our DesignDiffusion significantly outperforms the competitors. It obtains comparable, slightly worse than SDXL [34] in evaluating image aesthetic, *i.e.*, 40% v.s. 43%. It may be because adapting SDXL for the design image generation

task loses a bit of aesthetic.

**Automatic Layout Planning.** DesignDiffusion is a one-stage method without any layout planning module. We visualize examples compared with layout-dependent methods [5, 6] in Fig. 4. Samples from others often have an oversized font or dull layout design, while our DesignDiffusion generates images with curved, colorful, and diverse text. The results show that our DesignDiffusion has a strong automatic, flexible, and reasonable layout planning capability.

### 4.3. Ablation Studies

To assess the effectiveness of the proposed components within our framework, we conduct ablation studies. We explore various configurations, with the baseline being the original SDXL model. We denote fine-tuning on our dataset as “FT”, the utilization of prompt enhancement,  $\mathcal{L}_{char}$ , and self-play DPO as “PE”, “ $\mathcal{L}_{char}$ ” and “SP-DPO”, respectively. The ablation results are summarized in Tab. 3, revealing the following notable insights.

**Fine-tuning enhances the quality of design image generation:** Fine-tuning SDXL with design images results in a significant improvement in FID. Despite this enhancement in image quality, achieving satisfactory accuracy in rendered text remains a challenge for existing diffusion models, even with extensive fine-tuning on large-scale datasets.

**Effect of prompt enhancement:** Our investigation highlights limitations of current Byte Pair Encoding (BPE) tokenization, particularly in providing precise character-level cues for the visual text rendering task. Incorporating prompt enhancement leads to a significant improvement across various metrics, notably enhancing visual text accuracy because of the introduced character-level information.

**Localized attention through  $\mathcal{L}_{char}$ :** By utilizing character segmentation masks to guide the cross-attention maps within

Table 3. Ablation studies of the proposed framework. **Bold** and underline denotes the first and second best performance. “FT” and “PE” represent whether to apply fine-tuning and prompt enhancement, respectively.

| FT | PE | $\mathcal{L}_{char}$ | SP-DPO | FID↓         | Text Precision↑ | Text Recall↑ | Text F-score↑ | Text Accuracy↑ |
|----|----|----------------------|--------|--------------|-----------------|--------------|---------------|----------------|
| ×  | ×  | ×                    | ×      | 45.10        | 0.517           | 0.461        | 0.488         | 0.241          |
| ✓  | ×  | ×                    | ×      | 22.73        | 0.715           | 0.678        | 0.696         | 0.388          |
| ✓  | ✓  | ×                    | ×      | 19.89        | 0.862           | 0.799        | 0.829         | 0.572          |
| ✓  | ✓  | ✓                    | ×      | <b>19.29</b> | <u>0.881</u>    | <u>0.831</u> | <u>0.855</u>  | <u>0.616</u>   |
| ✓  | ✓  | ✓                    | ✓      | <u>19.87</u> | <b>0.888</b>    | <b>0.837</b> | <b>0.862</b>  | <b>0.631</b>   |



Figure 4. Visual comparisons of the capabilities of automatic text layout planning from our model with those of planning by language models. Upon training, our DesignDiffusion has demonstrated the ability to generate flexible and well-organized text layouts effectively.

the UNet architecture, our  $\mathcal{L}_{char}$  loss facilitates focused attention on text regions. With  $\mathcal{L}_{char}$ , we observe a further enhancement in text precision, recall, F-score, and accuracy, emphasizing the effectiveness of localizing cross-attention maps using character segmentation masks.

**Enhanced visual text accuracy with SP-DPO:** Subsequent to fine-tuning, we integrate SP-DPO to further improve the generation quality. SP-DPO significantly enhances text quality across all text metrics. Regarding the slight decline in the FID score, this can be attributed to the specific implementation of SP-DPO, which relies heavily on the selection of winning and losing data. Currently, the winning-losing pairs are determined based on the text accuracy. If we incorporate image quality into the criteria for constructing the losing data, we anticipate that the FID score could improve further. Notably, our current approach does not visually compromise image quality while enhancing text quality.

## 5. Discussion and Conclusion

In this paper, we introduce DesignDiffusion, a diffusion-based fine-tuning framework designed for synthesizing design images from textual descriptions. Unlike previous approaches that handle visual and textual elements independently, potentially limiting creativity and leading to style or color inconsistencies in style or color harmony, DesignDiffusion offers an end-to-end, one-stage diffusion-based solution. We explore including a distinctive character embedding and a character localization loss for enhancing visual text learning. Additionally, we employ an SP-DPO fine-tuning strategy to further improve the generation quality. Our extensive experiments demonstrate the effectiveness and robustness of DesignDiffusion in addressing the complexities of design image generation. We anticipate that our research will stimulate further exploration in design image generation, fostering innovation and advancement in this field.

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